

## **CONSUMABLES, REPLACEABLE COMPONENTS, AND SUBSCRIPTION HARDWARE EMBODIMENTS**

In various embodiments, the intraluminal compliance and distance profiling device is supported by consumable components and replaceable modules configured to improve hygiene, maintain performance, and enable scalable product offerings. Consumables and replaceable components may be used to support recurring maintenance, periodic replacement schedules, or tiered service models.

Consumable embodiments may include single-use or limited-use sleeves, distal tip covers, inflatable cuff covers, sealing films, lubricant-compatible barrier layers, or sterile packaging components. Consumables may provide hygienic separation, protect sensing surfaces, and reduce contamination risk during repeated operation.

In certain embodiments, functional components of the device are replaceable, including inflatable cuffs, pressure sensing sleeves, distal sensing windows, distal tip modules, valves, seals, and coupling membranes. Replaceable modules may be user-installed or service-installed and may extend device longevity while maintaining measurement reliability.

The accessory ecosystem may include refillable or replaceable pumping and inflation components such as cartridges, micro-pumps, filters, valves, or pressure equalization elements. In certain configurations, consumables may include pre-calibrated inflatable modules enabling standardized compliance measurement without recalibration by the user.

Replaceable or upgradable electronic modules may also be provided, including communication modules, batteries, sensor cartridges, or processor modules. Modular electronics enable future upgrades, performance enhancements, or regulatory compliance modifications without redesign of the entire device.

Subscription hardware embodiments may link access to consumable supplies, replaceable component shipments, calibration services, or model update services to recurring user plans. Software may track consumable usage count, replacement intervals, and maintenance compliance and may provide ordering prompts or automated fulfillment options.

In certain embodiments, device operation is conditioned on confirmation that approved consumables are installed or that replacement schedules are satisfied. Such conditioning may be implemented through physical identification features, software confirmations, secure pairing protocols, or usage tracking mechanisms.

Consumable and replaceable components may incorporate identifiers such as barcodes, RFID, NFC tags, cryptographic tokens, or machine-readable markings to support authentication, lot tracking, and quality assurance. Authentication systems may further support counterfeit prevention and controlled supply chain governance.

Accordingly, consumable, replaceable, and subscription hardware embodiments provide mechanisms to maintain safety and hygiene, preserve measurement accuracy, enable modular upgrades, and support scalable commercial deployment models while remaining within the functional scope of the disclosed intraluminal measurement platform.